

Riri on the floor: answering, reminding, and working a book

How Riri becomes ConnectDesk's call-centre agent. She answers the calls that today reach voicemail, knows who is calling and what they owe, tells them only after they prove who they are, books call backs on their own agent, and in time works a book of her own under the same dialler rules as every person on the floor. It is written against the "One Riri" plan of 11 September, so she stays one brain with one set of tools and guardrails, not a second assistant bolted onto the phone line. It also records what was found and fixed on Fastdial's first auto-dialler day, 29 September, because two of those findings decide how Riri must be built.

29 September 2026
Prepared

Birgen · BirgenAI
For

One Riri, 11 Sep 2026
Reads with

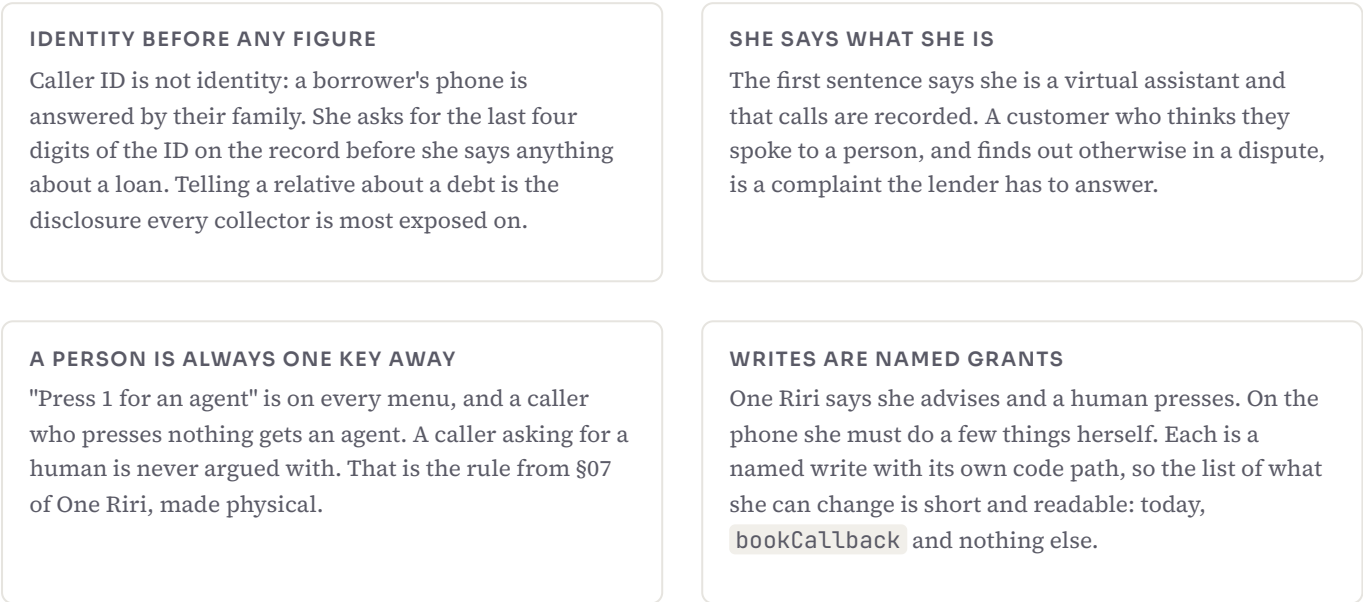
connected-suite
fafa1e7 to a522a8a · not deployed

01 Where it stands tonight

PIECE	STATE	WHAT THAT MEANS
Riri answers the inbound line (keypad menu)	Built · not deployed	Commit 7234ff8 . 1 an agent, with the caller's own agent first and by name · 2 the balance, after an ID check · 3 a call back, booked on the agent in CallBox. Hold with music instead of voicemail. 50 tests (npm run test:riri-line).
Riri's voice on the phone	Blocked	The Google key in this workstation's .env returns 403: requests to texttospeech.googleapis.com are blocked. The Cloud Text-to-Speech API has to be enabled on that key's project; production's key was not tested, and if it is the same key it is refused too. Until then the edge keeps the plain hunt, because it refuses to install a menu it cannot speak.
The media edge at Fastdial	Not installed	Everything on the inbound line runs on it. Today all 1,518 customer calls to Fastdial went to extension 716 and voicemail.
Riri working a book (reminders)	Designed	\$05. A virtual seat on the same dialler, the same contact policy, the same calling window.
Conversational Riri (speech in, speech out)	Designed	\$06. A voice gateway on the edge box, streaming speech-to-text and text-to-speech, and Riri Core with tools.
The "Riri Voice" tile on the handset	Corrected	It printed 90% of the floor's dials as "reminders placed today", a number nothing had done. It now says what she does and claims no count.
The four floor fixes from Lauryn's report	Built · not deployed	Appendix A. Settings that stick, outcomes that land on the call, calls that end, and no more ringing an agent who is mid-call.

02 One Riri, a new surface: the phone

The 11 September plan settled that there is one Riri with five contracts (host, context, corpus, tools, destination) and that every live number comes from a tool whose body checks tenancy and consent, never from a prompt. The phone is not an exception to that. It is a **surface**, **voice**, beside the console dock, the customer app, USSD and the partner embed. What changes on a phone line is not the brain but the rules of the room, and they are stricter than on a screen:



Two more rules carry over unchanged. She never states a figure she did not read from the record in this call: the balance is read when she speaks it, never cached. And she never reads back an ID number or a full phone number.

03 The architecture

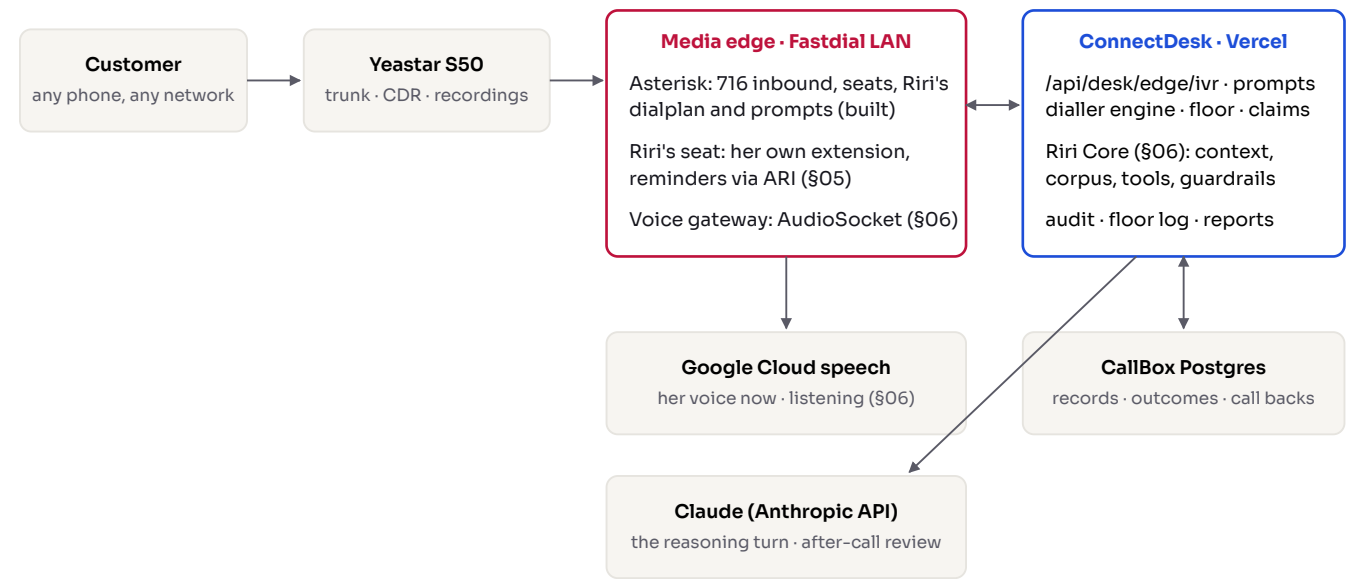


Figure 1 · Where each part of Riri runs. The edge holds the audio; ConnectDesk holds every decision and every fact.

PART	RUNS ON	WHY THERE
The call's audio	The media edge, on Fastdial's LAN	Vercel functions cannot hold a five-minute, two-way audio stream. The edge already carries the call and sits next to the PBX, so audio never crosses the office's internet line except to the speech services.
Every decision and every fact	ConnectDesk on Vercel	Who is calling, whose customer they are, whether the agent is free, whether the ID matched, what the balance is. One place, audited, testable without a phone. The edge only asks and plays.
Her voice	Google Cloud Text-to-Speech	The voice Riri already uses in the console (<code>en-GB-Neural2-C</code>), rendered at 8 kHz, the rate a phone call carries. Fixed prompts are rendered once and cached on the edge; only the balance is rendered per call, and deleted when the call ends.
Her reasoning (\$06)	Claude, through Riri Core	The same brain, tools and packs as every other surface. Per One Riri \$02, a second brain for the phone is the thing to avoid.
The book	CallBox (Fastdial) · CollectBox and the LMS (Micromart)	Read through the tools that already exist (<code>lookupCaller</code> , <code>callerAccounts</code> , <code>getRecord</code>) and written only through named grants.

04 Three levels of Riri, built in this order

LEVEL	WHAT THE CALLER GETS	WHY THIS ORDER
1 • Keypad Riri built	A menu in her voice; their own agent by name; hold, not voicemail; the balance after an ID check; a call back booked on their agent.	No speech recognition and no model on the call, so nothing she says can be invented, and it works on every phone. It turns 1,518 voicemails a day into answered calls or booked call backs.
2 • Riri on a book next	Reminder calls on the courtesy books (PRE, DUE): right-party check, the instalment and date, 1 an agent now, 2 I have paid, 3 call me back.	The dialler, claims and contact policy already exist. She becomes a seat on them rather than a system of her own, so nobody is over-called because a machine dialled them.
3 • Conversational Riri after 2	She understands speech in English and Swahili, answers questions about their loan from the record and the lender's pack, takes a promise with a read-back, and hands over with a summary.	It needs an evaluation set of real calls to be safe, and levels 1 and 2 produce exactly those calls, recorded, with outcomes.

05 Allocating Riri to a book

"Put Riri on the PRE book" should mean the same click a supervisor makes for Calvin. She is a **seat**:

- **A seat flagged virtual**, with an extension of her own on the PBX (for example 750, registered by the edge) and a **CallBox user named Riri**, created by Fastdial's CallBox administrator the way *#87 SMS FAST DIAL* exists. Outcomes she saves then carry her name in the history, the reports and the client reports, and the PBX records and bills her calls like anyone's.
- **Assigned like a person**. In the command centre she is moved to a campaign or bucket and given a mode. The dialler's `nextForSeat` hands her customers with the same claims (no two seats get one customer), the same calling window, the same attempts per day, and the same weekly contact policy. Her calls count toward the caps for everyone.

- **A worker on the edge** asks for her next customer, originates the call through the PBX as her extension, and runs the script. Answering-machine detection (Asterisk's `AMD`) hangs up on voicemail: a message about a debt left on a shared voicemail is a disclosure.
- **A line cap.** The S50 carries 25 calls. Riri gets a setting for at most N lines (start at 3), so she never takes the lines the people need.

The reminder, level 2 (keypad)

"Hello, this is Riri, a virtual assistant calling from Fast Dial on behalf of Fourth Generation Capital. I have a message for Jane. If this is Jane, press 1." → ID check → "Your instalment of 4,000 shillings is due on Friday the 2nd. You can pay through M-Pesa, paybill 992701. To speak to an agent now, press 1. If you have already paid, press 2. To have us call you back, press 3."

- 1 transfers into the hunt, the account's own agent first, and their screen opens on the case with Riri's note: *"Reminder delivered, customer asked for an agent."*
- 2 saves the outcome "Says paid". The contact policy's "quiet after paying" then holds them back once the payment lands.
- 3 books the call back on their agent.
- No key, a wrong ID, or not the right person: a neutral goodbye with no figures, and the outcome "Not right party".

Which books she may work

BOOK	RIRI?	WHY
PRE / DUE (not yet late)	Yes, first	Courtesy reminders: high volume, low judgement, the calls agents least want and customers least mind.
Watch 1 (a few days late)	Later	Once level 2 has shown right-party and payment rates on PRE/DUE, and only with "press 1 for an agent" first.
Watch 2-3, NPL, legal, disputes	No	Negotiation, hardship and disputes need a person. Stop outcomes (wrong number, deceased, disputed) already keep the dialler away; they apply to her too.

Measured the way people are: right-party contacts, promises, and payments within three days, using the same "kept" rule the client reports use (`lib/desk/report-rules.ts`), plus how many asked for an agent and how many opted out.

06 Conversational Riri: the infrastructure

STAGE	COMPONENT	NOTES
Audio out of the call	Asterisk <code>AudioSocket</code> on the edge	Streams the call's audio (8 kHz signed linear) over a TCP socket to a local service and plays back whatever it sends. Simpler than RTP external media, and it stays on the box. Check the module first: <code>module show like audiosocket</code> .
Voice gateway	A Node service on the edge box	Holds the call's session. Runs voice-activity detection and end-of-turn detection, and handles barge-in: the caller talking over Riri stops her mid-sentence. It is the only long-running part, and the reason it cannot be on Vercel.

Speech to text	Google Cloud Speech-to-Text, streaming, <code>en-KE</code> and <code>sw-KE</code>	Same vendor and billing as her voice. Code-switching and Sheng are the weak point: to be measured on real calls, and the keypad stays available on every turn.
The turn	Riri Core over HTTP, <code>surface: "voice"</code>	The same context descriptor, tier A and B corpus and tools as every other surface, with the verified caller as the subject. Tools are listed below.
Text to speech	Google Cloud Text-to-Speech, streamed by sentence	Her first sentence plays while the rest is still being written.
After the call	Summary, promise extraction, QA, compliance	Asynchronous, off the call's latency path (below). It absorbs today's AI call review, which runs on Gemini.

The latency budget

A caller hears a pause of more than about a second and a half as the line going dead. The target is **first audio within 1.5 s of the caller finishing**, split roughly into end-of-turn detection (~0.3 s), the final transcript (~0.3 s), the model's first sentence (~0.6 s) and the first synthesised audio (~0.3 s). These are budgets to measure against, not measurements. When a turn needs a tool (a balance read), she says a short holding line ("Let me check that for you") while it runs.

The model

USE	MODEL AND SETTINGS	REASON
The live turn	<code>claude-opus-5</code> , adaptive thinking, <code>effort Low</code> , streaming	The most capable current model, at the effort level for latency-sensitive routes. On the newest models, lower effort usually holds quality better than dropping to an older model. If measurements miss the budget, fast mode (Claude API only, at a premium) is the next lever.
After-call review	<code>claude-opus-5</code> , <code>effort high</code> , through the Message Batches API	Not latency-bound. Batches run asynchronously at half the price. This is where summaries, promises, QA scores and compliance flags come from.
A cheaper model	<code>claude-sonnet-5</code> or <code>claude-haiku-4-5</code>	Only once an evaluation built from level 1 and 2 calls shows quality holds. Your decision on cost, not a default.

Every request checks `stop_reason` before reading the answer, and uses the API's server-side fallback on a refusal, so a refused turn becomes "let me get you to an agent" rather than silence. List price for `claude-opus-5` is \$5 / \$25 per million input / output tokens (Anthropic's published table, June 2026). Re-check before budgeting.

Why not a speech-to-speech model

Native audio models reply faster because they skip the text step. But the model would then decide what to say outside Riri Core, and every consent check, tool boundary and pack source would have to be rebuilt around it: the four-Riris problem the 11 September plan exists to end. It stays an option to measure against, but only if it calls the same tool endpoints and nothing else.

Tools on the phone

TOOL	KIND	CHECKED INSIDE THE TOOL
<code>caller.identify(msisdn)</code>	read	Org scope. Returns candidates, never figures.
<code>caller.verify(record, idLast4)</code>	read	Built as <code>ivrBalance</code> : audited; four misses an hour lock the number.

account.balance · account.schedule · account.promises	read	A verified caller in this call. Figures are read, not remembered.
callback.book(record, when)	write grant	Built as bookCallback : only a task on the account's own agent, inside office hours.
ptp.record(record, amount, date)	write grant	Read back and confirmed on the keypad ("press 1 to confirm") before it is saved.
handoff.agent(owner any, summary)	action	Always available. The summary pops on the agent's screen before they speak.
disposition.save(record, outcome)	write grant	As the CallBox user Riri, from a short list of outcomes she may use.
sms.paybill(record)	write grant	blocked Fastdial has no SMS provider (checked 28 Sep).

07 Data, consent and the lenders

- **The lenders are the controllers** of their borrowers' data and Fastdial processes it for them. Riri adds two sub-processors, Google (speech) and Anthropic (reasoning). Each client lender (4G Capital, Mular Credit and the rest) should agree in writing to Riri stating balances on its behalf and to the sub-processors, before the balance option is switched on for their contracts.
- **Recording and disclosure** are in the first sentence. Every balance disclosure and every failed ID check is an audit row (desk.riri.balance, desk.riri.id-mismatch). The per-call balance audio is deleted when the call ends.
- **No training on the book.** Per One Riri \$08, training on a member's data is only under model.train consent and de-identified. Evaluation sets built from recorded calls follow the same rule.
- **Coverage today:** 9,938 of the 12,608 records on the three pilot contracts (79%) carry an ID number she can check. The rest are put through to an agent for the balance.

08 The build

#	DELIVERABLE	DONE WHEN
A	Keypad Riri on the inbound line built	The edge is installed, Text-to-Speech is enabled and the box is ticked. Two test phones: the menu plays; press 1 while the owner is on a call and hear their name; a wrong ID then the right one; a call back appears under the agent and in the floor log.
B	Riri as a seat on PRE/DUE (keypad reminders)	Her seat, extension and CallBox user exist; the line cap is set; a reminder campaign runs inside the calling window; outcomes appear under "Riri"; "press 1" pops the case on an agent's screen.
C	The voice gateway and Riri Core's voice host	On an evaluation set of real recorded calls (consented, de-identified), her answers cite only figures the tools returned, and first audio is within budget at the 90th percentile.
D	Conversational inbound, then conversational reminders	C holds in a monitored pilot on one contract, with a supervisor listening in and a kill switch per floor and per book.
E	Swahili and code-switching	Measured separately. The keypad remains the fallback on every turn.

09 Decisions and switches that are yours

ITEM	WHY IT WAITS ON YOU
Enable the Cloud Text-to-Speech API on the Google key's project	The blocker for A. The key exists and is refused today.
Install the media edge at Fastdial, and set 716's password	A small Ubuntu box on their LAN (<code>bridge/edge/README.md</code>). Nothing on the inbound line works without it.
What callers hear her called	"Riri", or white-labelled per host ("Fast Dial's assistant"). One Riri recommends Riri as the product name, white-labelled per host.
Each lender's agreement to balance disclosure and the sub-processors	\$07. Until then, leave option 2 out of the menu for that lender's contracts.
The books she may work, and her line cap	\$05. PRE/DUE and 3 lines is the suggested start.
A CallBox user named Riri	Created by Fastdial's CallBox administrator. Needed for B.
An SMS provider for Fastdial	For paybill-by-SMS and reminders by text. None exists today.
Deploying tonight's commits	Pushing <code>main</code> deploys production. The four floor fixes (Appendix A) matter to Lauryn tomorrow morning.

A Lauryn's report, 29 September: what was found

From Fastdial's first auto-dialler morning. The findings are from ConnectDesk's own call ledger and CallBox's call records for the day (read-only).

SHE REPORTED	WHAT THE RECORDS SHOW	FIXED IN
"I need the right to set the auto-dialler time"	She has the right (Org Admin). The settings page threw every unsaved edit away each time the live floor refreshed, every few seconds. And a save went out as three writes that could land over each other; the floor's 15-second tick could also put the old settings back.	fafe1e7 · one save action, a draft that holds, and the floor row writes only what changed
"Incoming calls come through while an agent is on a call, and the ongoing call drops"	Not customers: all 1,518 inbound calls went to 716 and voicemail. It was ConnectDesk. Every call rings the agent's phone first, and 10 dialler calls rang agents mid-conversation (Edith, 700: 10:39:56, 10:40:57, 10:42:11, inside one call from 10:37:17 to 10:42:55). Only 61 of 506 calls were ever matched as answered, and the two-minute rule ended 30 calls while they were still going on. The seat went Ready and the dialler rang again.	a522a8a · the PBX's extension state decides: no dialler or click-to-call onto a busy extension, and no call ended by the clock while its extension is on it
"Calls end on the iPhone but still show as active, and the agent cannot end them"	Nothing stayed open in the database: the call record arrived a median 17 s, at most 47 s, after the real end. But an answered call with no record could wait forever. And pressing End could be undone by a status check already in flight, which put the call back on screen.	d30764e · End stops the check first; an answered call gone from the PBX ends 20 s later
"A submitted disposition does not show in the customer's history"	39 answered calls had no outcome on the call. 14 (all in the 10:33–10:56 dialler test) were never saved: the next call opened 36–47 s later and replaced the case file being typed into. 23 were saved into a second call-log row, so the history showed the call with no outcome and the outcome on a line of its own.	d30764e · the dialler waits for the outcome after an answered call; an outcome without its call's row completes that row

Also measured: 67 dialler calls in progressive mode; about 20% of each agent's calls were dialled outside ConnectDesk (CallBox or the phone). ConnectDesk now sees those through the extension state too.

B Listen, whisper and barge: the PBX settings

ConnectDesk asks the Yeastar to join the supervisor to the call; the Yeastar decides whether that is allowed. The message Lauryn saw, *"That extension does not allow being monitored"*, is the PBX's answer for Calvin's extension (744).

1. Sign in to the Yeastar web interface as an administrator, from the office: `http://192.168.1.150:8088`. Menu names vary slightly between firmware versions.
2. **Agents:** PBX Settings → Extensions → edit each agent's extension → **Features** → tick **Allow Being Monitored** → Save. If the firmware offers *Bulk Edit* on the Extensions list, select every agent extension and set it once.

3. **Supervisors:** edit Lauryn's extension (749), and Sammy's and Fred's when they have one → Features → **Monitor Mode: Extensive** (listen, whisper and barge-in) → Save.
4. **Apply** the configuration if the PBX asks.
5. Lauryn's own extension must be signed in on a phone: ConnectDesk rings it to join her to the call. In the command centre, open an agent's seat on a call → Listen, Whisper or Barge → answer her phone.

Also worth setting while there: on each agent extension, Features → untick **Call Waiting**. A second call to a busy extension then gets a busy signal instead of ringing over the customer. ConnectDesk no longer sends one (`a522a8a`), but this makes sure a call from anywhere else cannot either.